

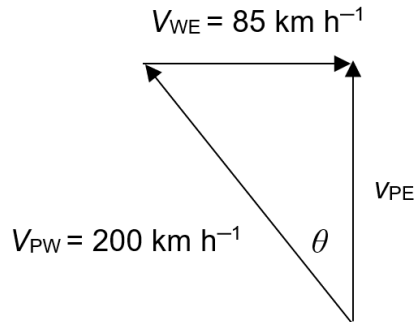
- 4 The speed of an aeroplane in still air is 200 km h^{-1} . The wind pushes it from the west at a speed of 85.0 km h^{-1} .

In which direction must the pilot steer in order to fly due north?

- A 23.0° east of north
 B 23.0° west of north
 C 25.2° east of north
 D 25.2° west of north

Ans: D

$V_{PE} = V_{PW} + V_{WE}$ where V_{PE} is velocity of plane relative to Earth
 V_{PW} is velocity of plane relative to Wind
 V_{WE} is velocity of wind relative to Earth



$$\sin \theta = 85/200$$

$$\theta = 25.2^\circ \text{ west of north}$$

- 5 A radio-controlled toy car travels along a straight line for a time of 15 s